

Contents

EEE 554 — Lecture #8	1
Multiple RVs, cont'd	1
Abbreviated notation	1
Probability of an event in $\tilde{\mathbf{X}}$	1
Conditioning on an event	2
Example	2
Independent RVs	2
Example: independence via marginalization of the joint PDF	2
(a)	2
(b)	3
Expectation rule	3

EEE 554 — Lecture #8

Multiple RVs, cont'd

Abbreviated notation

(a) Let $\tilde{\mathbf{X}}$ denote the n -tuple $(X_1, \dots, X_n)$ of random variables, and also the **random vector**:

$$\tilde{\mathbf{X}} = \begin{bmatrix} X_1 \\ \vdots \\ X_n \end{bmatrix}$$

(b) Let $\mathbf{x}$ denote the corresponding n -tuple (vector) of **deterministic variables**:

$$\mathbf{x} = (x_1, \dots, x_n)$$

(c) Abbreviated multiple-integral notation:

$$\int_A g(\mathbf{x}) d\mathbf{x} = \int \cdots \int_{(x_1, \dots, x_n) \in A} g(x_1, \dots, x_n) dx_1 \cdots dx_n$$

With this notation:

- the **joint CDF** of $X_1, \dots, X_n$ is $F_{\tilde{\mathbf{X}}}(\mathbf{x})$
- the **joint PDF** is $f_{\tilde{\mathbf{X}}}(\mathbf{x})$

Probability of an event in $\tilde{\mathbf{X}}$

For an event $A \subseteq \mathbb{R}^n$,

$$P(A) = P\{(x_1, \dots, x_n) \in A\} = \int_A f_{\tilde{\mathbf{X}}}(\mathbf{x}) d\mathbf{x}$$

Conditioning on an event

The conditional PDF of $\tilde{\mathbf{X}}$ given $\tilde{\mathbf{X}} \in A$ is

$$f_{\tilde{\mathbf{X}}}(\mathbf{x} | A) = \begin{cases} \frac{f_{\tilde{\mathbf{X}}}(\mathbf{x})}{P(A)}, & \mathbf{x} \in A, \\ 0, & \mathbf{x} \notin A. \end{cases}$$

Example

$$f_{\tilde{\mathbf{X}}}(x_1, x_2) = \begin{cases} 6x_1^2 x_2, & 0 < x_1 < 1, 0 < x_2 < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Let

$$A = \{x_2 > x_1^2\}$$

Then

$$P(A) = \int_0^1 \int_{x_1^2}^1 6x_1^2 x_2 dx_2 dx_1 = 6 \int_0^1 x_1^2 \left[\frac{x_2^2}{2} \right]_{x_1^2}^1 dx_1 = 3 \int_0^1 (x_1^2 - x_1^6) dx_1 = \frac{4}{5}$$

Thus,

$$f_{\tilde{\mathbf{X}}}(x_1, x_2 | A) = \begin{cases} \frac{6x_1^2 x_2}{4/5} = \frac{15}{2} x_1^2 x_2, & (x_1, x_2) \in A, \\ 0, & \text{otherwise.} \end{cases}$$

Independent RVs

Random variables $X_1, \dots, X_n$ are **statistically independent** if

$$f_{\tilde{\mathbf{X}}}(x_1, \dots, x_n) = f_{X_1}(x_1) \cdots f_{X_n}(x_n)$$

Example: independence via marginalization of the joint PDF

(a)

$$f_{\tilde{\mathbf{X}}}(x_1, x_2) = \begin{cases} \frac{3}{2} x_1^2 x_2, & 0 < x_1 < 1, 0 < x_2 < 2, \\ 0, & \text{otherwise.} \end{cases}$$

Marginals:

$$f_{X_1}(x_1) = \int_0^2 \frac{3}{2} x_1^2 x_2 dx_2 = \frac{3}{2} x_1^2 \left[\frac{x_2^2}{2} \right]_0^2 = 3x_1^2, \quad 0 < x_1 < 1$$

$$f_{X_2}(x_2) = \int_0^1 \frac{3}{2} x_1^2 x_2 dx_1 = \frac{3}{2} x_2 \left[\frac{x_1^3}{3} \right]_0^1 = \frac{1}{2} x_2, \quad 0 < x_2 < 2$$

Since $f_{X_1}(x_1)f_{X_2}(x_2) = f_{\tilde{\mathbf{X}}}(x_1, x_2)$, X_1 and X_2 are independent.

(b)

$$f_{\tilde{\mathbf{X}}}(x_1, x_2) = \begin{cases} 15x_1^2 x_2, & 0 < x_1 < 1, 0 < x_2 < 2 - 2x_1, x_1 + x_2 \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

Marginals (as written in the notes):

$$f_{X_1}(x_1) = \int_0^{2-2x_1} 15x_1^2 x_2 dx_2 = \frac{15}{2} x_1^2 (2 - 2x_1)^2 = 30x_1^2 (1 - x_1)^2, \quad 0 < x_1 < 1$$

$$f_{X_2}(x_2) = \int_0^{1-x_2/2} 15x_1^2 x_2 dx_1 = 5 \left(1 - \frac{x_2}{2}\right)^3 x_2, \quad 0 < x_2 < 2$$

Because $f_{X_1}(x_1)f_{X_2}(x_2) \neq f_{\tilde{\mathbf{X}}}(x_1, x_2)$, X_1 and X_2 are **not** independent.

Expectation rule

If $Z = g(X_1, \dots, X_n)$, then

$$\mathbb{E}[Z] = \int_{\mathbb{R}^n} g(\mathbf{x}) f_{\tilde{\mathbf{X}}}(\mathbf{x}) d\mathbf{x}$$

In particular:

(a) Moments of X_k :

$$\mathbb{E}[X_k^a] = \int_{\mathbb{R}^n} x_k^a f_{\tilde{\mathbf{X}}}(\mathbf{x}) d\mathbf{x}$$

(b) Linearity for linear combinations:

$$\begin{aligned} \mathbb{E}[a_1 X_1 + \dots + a_n X_n] &= \int_{\mathbb{R}^n} (a_1 x_1 + \dots + a_n x_n) f_{\tilde{\mathbf{X}}}(\mathbf{x}) d\mathbf{x} \\ &= a_1 \mathbb{E}[X_1] + \dots + a_n \mathbb{E}[X_n] \end{aligned}$$

(c) Vector form:

Define the expectation vector $\mu = \mathbb{E}[\tilde{\mathbf{X}}]$:

$$\mathbb{E}[\tilde{\mathbf{X}}] = \begin{bmatrix} \mathbb{E}[X_1] \\ \vdots \\ \mathbb{E}[X_n] \end{bmatrix} = \begin{bmatrix} \mu_1 \\ \vdots \\ \mu_n \end{bmatrix}$$

If A is an $m \times n$ matrix with entries a_{ij} , then the i -th component of the vector $A\tilde{\mathbf{X}}$ is $\sum_{j=1}^n a_{ij}X_j$. Taking the expectation:

$$\mathbb{E}[A\tilde{\mathbf{X}}] = \begin{bmatrix} \mathbb{E}[\sum_{j=1}^n a_{1j}X_j] \\ \vdots \\ \mathbb{E}[\sum_{j=1}^n a_{mj}X_j] \end{bmatrix} = \begin{bmatrix} a_{11}\mu_1 + \cdots + a_{1n}\mu_n \\ \vdots \\ a_{m1}\mu_1 + \cdots + a_{mn}\mu_n \end{bmatrix} = A\mathbb{E}[\tilde{\mathbf{X}}]$$

Similarly, for the transpose (row vector form):

$$\mathbb{E}[\tilde{\mathbf{X}}^T A^T] = \mathbb{E}[\tilde{\mathbf{X}}^T] A^T$$